

1.14 GRADIENT ECHO (GRE) FORMATION

Gradient echoes are formed by gradient reversal, not 180° RF refocusing, and are sensitive to T_2^* effects.

WHAT IS A GRADIENT ECHO?

A small flip angle RF pulse tips M_z into the transverse plane. Dephasing occurs due to spin-spin interactions and magnetic field inhomogeneities (ΔB_0).

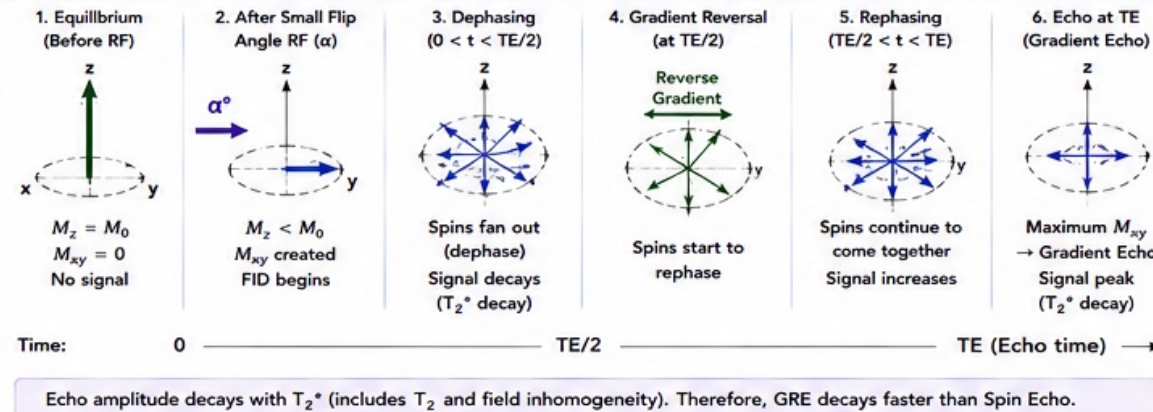
A gradient reversal rephases spins that were dephased by the applied gradient, forming an echo at time TE.

Because static field inhomogeneity is NOT refocused, GRE signal decays with T_2^* (faster than T_2).

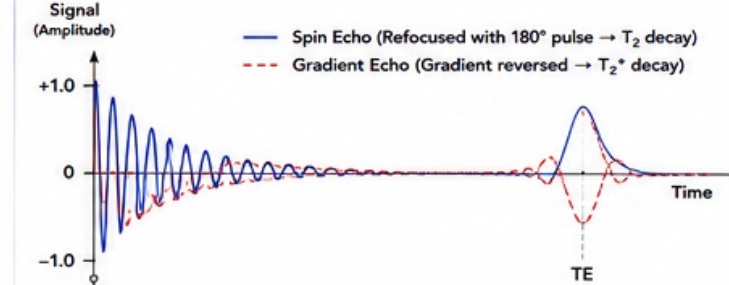
KEY DIFFERENCE

- Spin Echo:** 180° RF pulse refocuses spins → removes static ΔB_0 effects → T_2 decay.
Gradient Echo: Gradient reversal only → does NOT remove static ΔB_0 → T_2^* decay.

GRE TIMELINE AND MAGNETIZATION BEHAVIOR



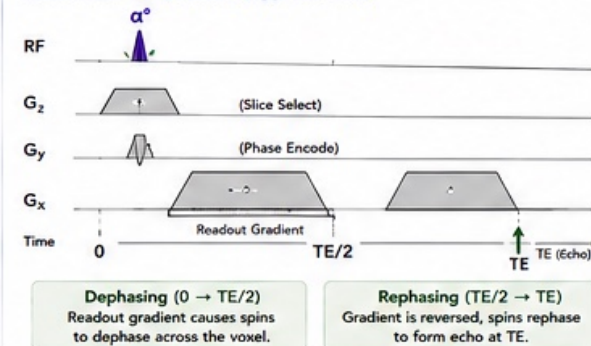
SIGNAL COMPARISON: SPIN ECHO vs GRADIENT ECHO



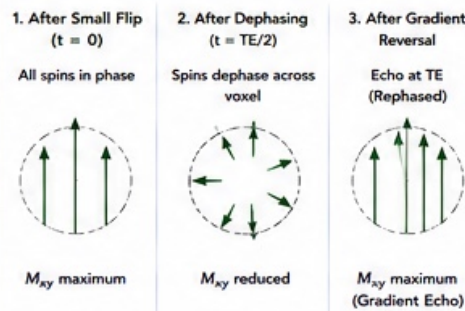
Key Points

- GRE decays faster because it includes additional dephasing from static field inhomogeneity.
- Spin Echo always has higher signal at long TE (in homogeneous tissue).
- In areas of susceptibility (air-tissue, metal, blood), GRE shows more signal loss.

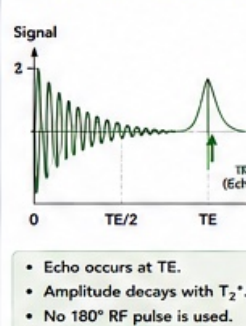
GRE PULSE SEQUENCE (Typical GRE)



DEPHASING AND REPHASING (IN-PHASE LINE)



ECHO FORMATION (GRE)



TYPICAL T_2^* VALUES AT 1.5T (APPROX.)

Tissue	T_2^* (ms)
Liver	10 – 20
Gray Matter	15 – 30
White Matter	20 – 40
Skeletal Muscle	15 – 30
Fat	20 – 40
CSF	40 – 60
Myocardium	10 – 20
Iron (hemosiderin)	1 – 5
Air / Lung	< 1

Values vary with field strength, sequence, oxygenation, and subject.

CLINICAL RELEVANCE (Why GRE is Important)

- Fast imaging (no 180° pulse → shorter TR/TE).
- Highly sensitive to susceptibility (ΔB_0) effects.
- Used in:
 - T_2^* -weighted imaging
 - Functional MRI (BOLD fMRI)
 - SWI (Susceptibility Weighted Imaging)
 - Dynamic contrast (FLASH)
 - MRA (TOF, contrast-enhanced)
 - Metal artifact evaluation
 - Field mapping (phase images)

Example: SWI (Brain)

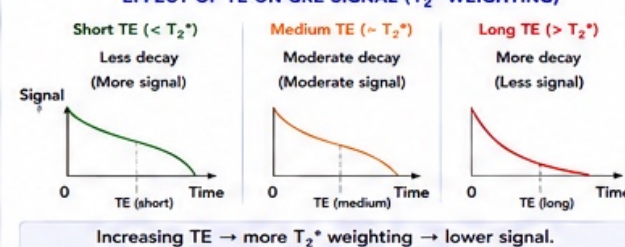


GRE (T_2^*) accentuates venous structures and microbleeds.

SPIN ECHO vs GRADIENT ECHO – KEY COMPARISON

Feature	Spin Echo (SE)	Gradient Echo (GRE)
Refocusing Mechanism	180° RF pulse	Gradient reversal
Refocuses Static ΔB_0 ?	Yes	No
Decay Constant	T_2	T_2^* ($\leq T_2$)
Signal at Long TE	Higher	Lower
Susceptibility Sensitivity	Low	High
SAR	Higher	Lower
Speed	Slower	Faster
Common Sequences	SE, TSE	GRE, FLASH, SPGR

EFFECT OF TE ON GRE SIGNAL (T_2^* WEIGHTING)



COMMON MISCONCEPTION

"GRE and Spin Echo are the same."
Reality: GRE does not use a 180° pulse. It cannot refocus static field inhomogeneity, so it always decays with T_2^* .

T_2^* = combined effect of T_2 (spin-spin relaxation) + field inhomogeneity (ΔB_0 variations).

KEY TAKEAWAY

- GRE uses gradient reversal to form an echo.
- It is sensitive to T_2^* (includes field inhomogeneity).
- GRE decays faster than Spin Echo.
- Smaller flip angle → more M_z left → more steady-state signal.
- GRE is the basis for fast imaging and susceptibility-sensitive MRI.

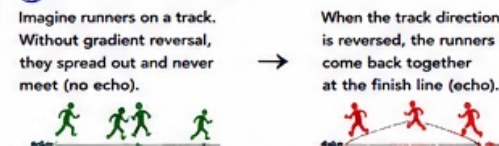
ENGINEERING INSIGHT

- GRE signal is highly sensitive to:
 - Shimming (ΔB_0 uniformity)
 - Susceptibility sources (air, metal, implants)
 - Gradient timing and amplitude accuracy
 - Eddy currents and gradient distortion
- Poor shim or environment → rapid signal loss.

FIELD SERVICE TIP

- Verify gradient delay calibration (especially readout gradient) for proper echo formation.
- Check shim quality before advanced sequences (e.g., SWI, fMRI).
- Inspect RF cables, preamp, and coil tuning for optimum SNR.
- Monitor room for ferromagnetic objects!

REAL-WORLD ANALOGY



SYSTEM IMPACT

GRE's sensitivity to ΔB_0 makes it ideal for detecting small magnetic disturbances but also vulnerable to:

- Motion (off-resonance changes)
- Metal (large local ΔB_0 variations)
- Poor shim

Result: Signal loss, distortion, or artifacts.

QUICK REFERENCE

Echo Time (TE)	Time of echo peak
Mechanism	Gradient reversal
Decay Constant	T_2^*
Refocuses Static ΔB_0 ?	No
Typical Use	Fast, susceptibility-sensitive imaging